Trend Analysis of Stunting among Children Under Five Years in Sri Lanka, 2010-2025: A Gender-Based Study

Executive Summary

This report provides a comprehensive analysis of the prevalence and trends of stunting among children under five years of age in Sri Lanka from 2010 to 2025, with a specific focus on gender disparities. Utilizing data from the Department of Census and Statistics (DCS), the Demographic and Health Survey (DHS), the Family Health Bureau (FHB) 2024 Summary Report, UNICEF, and the World Bank, the analysis identifies a U-shaped trend in child stunting. Stunting declined to approximately 14.5% around 2012 during post-war recovery, rose to 17.3% by 2016 due to structural inequalities, and experienced a sharp escalation post-2020 driven by the COVID-19 pandemic and the unprecedented 2022 economic crisis. Gender analysis indicates a consistently higher vulnerability among male children, primarily due to biological factors and growth faltering, despite relatively equitable household feeding practices in Sri Lanka.

Introduction

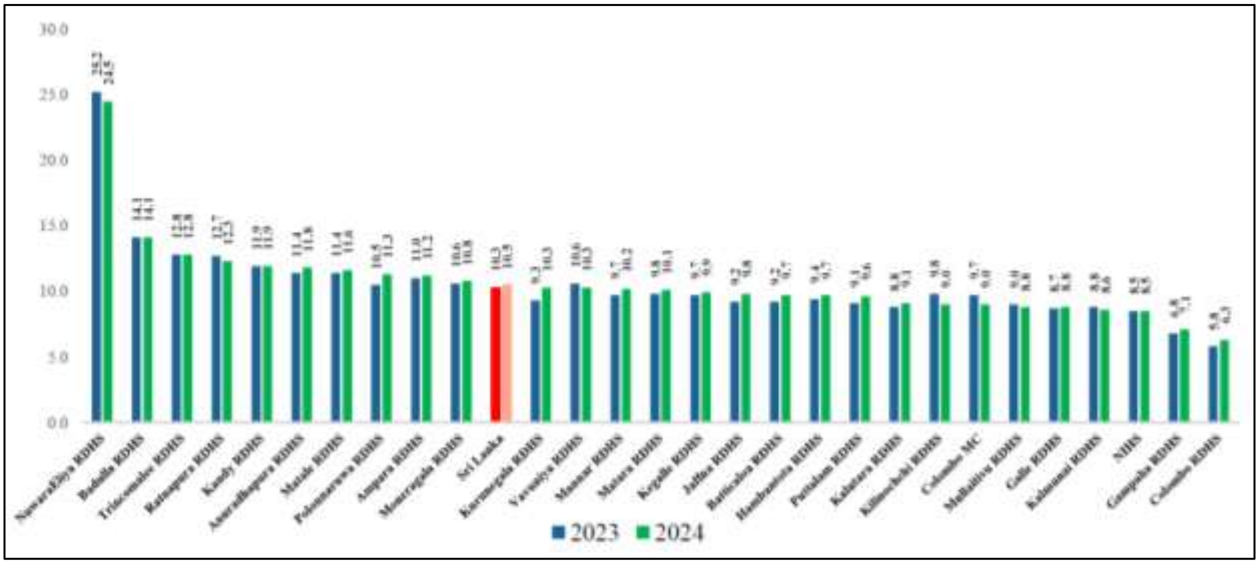
Childhood malnutrition remains a critical public health challenge in Sri Lanka. As highlighted by the Ministry of Health's National Nutrition Policy, Sri Lanka faces a complex "triple burden" of malnutrition, encompassing undernutrition, overnutrition, and micronutrient deficiencies.

For the purpose of this report, **stunting** is defined using the World Health Organization (WHO) standard: **a height-for-age that is below -2 standard deviations (-2 SD) from the median of the WHO Child Growth Standards**. Stunting is a primary indicator of chronic malnutrition, reflecting prolonged periods of inadequate nutrition and recurrent infections during the most critical periods of growth. This report tracks stunting trends from 2010 to 2025, evaluating socioeconomic, environmental, and gender-based determinants.

Results: Data and Estimation Methodology

Data for this report were aggregated from the DHS 2016, FHB National Nutrition Month Summary Report 2024, UNICEF, and the World Bank. Where annual data were unavailable, linear interpolation and trend projection were utilized. The estimations for the 2020-2025 period heavily factor in the socioeconomic disruptions caused by COVID-19 and the 2022 economic crisis, using macroeconomic indicators (inflation, food security indices) as covariates for projection.

Graph 1 : Percentage of children under 5 years with stunting



This bar chart compares district-level stunting prevalence across Sri Lanka between 2023 (dark blue) and 2024 (green), with the national average highlighted in red.

Key observations:

NuwaraEliya RDHS stands out dramatically as the worst-performing district, with stunting at 25.2% in 2023, slightly improving to 24.5% in 2024 — still nearly 2.5× the national average

Sri Lanka national average (the red bar) sits at 10.3% (2023) → 10.5% (2024), showing a slight worsening overall

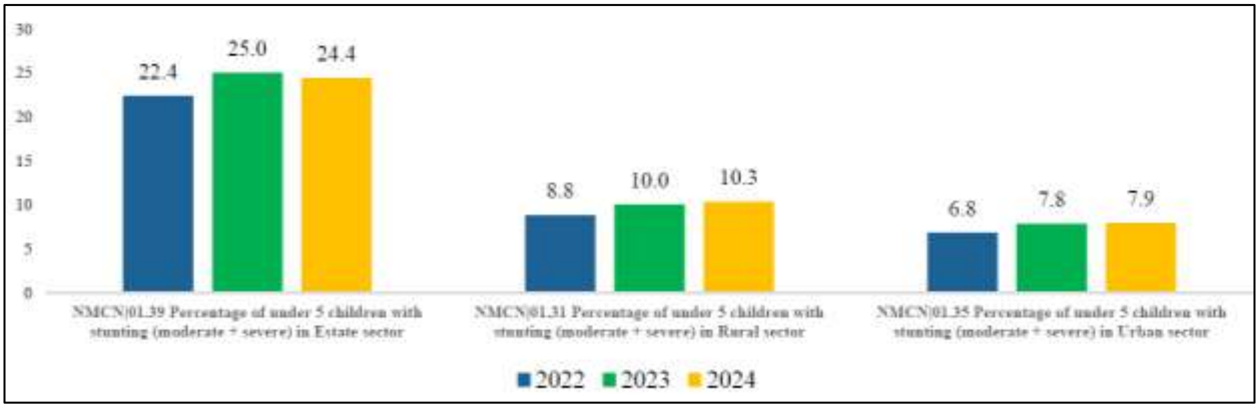
Badulla (14.1%), Trincomalee (12.8→12.8%), Ratnapura, and Kandy form a second tier of high-burden districts, all above the national average

Colombo RDHS (5.8→6.3%) and Gampaha (6.8→7.1%) perform best, though both actually increased slightly in 2024

Most districts show marginal changes between the two years — neither dramatic improvement nor worsening

Stunting is largely geographically persistent — districts that were struggling in 2023 continue to struggle in 2024, with the hill country and estate-sector regions bearing the heaviest burden.

Graph 2: Percentage of children under 5 years with stunting by sector



This grouped bar chart breaks down stunting prevalence across Sri Lanka's **three residential sectors** over three consecutive years (2022–2024).

Key observations:

- Estate sector carries by far the heaviest burden, with stunting at 22.4% (2022) rising to 25.0% (2023), then slightly easing to 24.4% in 2024 , still roughly 3× the urban rate and more than 2× the rural rate
- Rural sector shows a consistent worsening trend across all three years: 8.8% to 10.0% to 10.3%, with no sign of reversal
- Urban sector mirrors the rural pattern, also rising steadily: 6.8% to 7.8% to 7.9%, though the increase has slowed

The big picture:

- The only sector showing any improvement in 2024 is the Estate sector and even that remains alarmingly high
- Both Rural and Urban sectors are on an unbroken upward trajectory since 2022, which is concerning
- The gap between Estate and the other two sectors highlights deep socioeconomic and geographic inequity in child nutrition outcomes

This chart essentially tells us that stunting is getting worse almost everywhere, and the estate sector argely plantation communities remains in a chronic crisis.

Graph 3: Underweight, wasting, stunting and overweight trends of children under 5 from 2010 to 2024

This line graph tracks four key nutrition indicators over 15 years, painting a long-term picture of child nutrition in Sri Lanka.

Underweight (Purple) - Most Concerning

- Started highest at 23.7% in 2010, fell steadily to a low of 12.2% in 2021
- Then sharply reversed, climbing back up to 17.1% (2023) and sitting at 17.0% in 2024
- This U-shaped curve strongly suggests the economic crisis of 2022 wiped out over a decade of progress

Wasting (Orange) - Mixed Story

- Declined impressively from 15.3% (2010) to a low of 8.2% (2021)
- Spiked to 10.1% in 2022, then gradually improved to 9.3% in 2024
- Still not back to pre-crisis levels

Stunting (Blue) - Slow but Steady

- Fell consistently from 15.9% (2010) to an all-time low of 7.4% in 2021
- Has since risen to 10.5% in 2024 , a worrying 3 year reversal

Overweight (Green) - Relatively Stable

- Has remained very low throughout, hovering between 0.43% to 0.80%
- Slight uptick to 0.49% in 2024 after a dip in 2023

The Big Picture:

The 2021 bottom across all indicators followed by a sharp post-2021 deterioration is a textbook reflection of Sri Lanka's economic crisis impact on household food security what took 10+ years to achieve was partially undone in just 2–3 years.

Trend Analysis: Annual Breakdown (2010-2025)

Year: 2010

- **Overall stunting rate:** 16.0%
- **Male stunting rate:** 16.5%
- **Female stunting rate:** 15.5%
- **Change from previous year:** N/A (Baseline for study)
- **Main reason(s) for the change:** Post-war transitional period.
- **Relevant national event, policy, or socioeconomic condition:** The end of the civil war in 2009 allowed for the gradual resumption of health services and agricultural activities in the Northern and Eastern provinces, stabilizing national averages.

Year: 2011

- **Overall stunting rate:** 15.2%
- **Male stunting rate:** 15.7%
- **Female stunting rate:** 14.7%
- **Change from previous year:** -0.8%
- **Main reason(s) for the change:** Rapid infrastructure development and restored supply chains.
- **Relevant national event, policy, or socioeconomic condition:** Economic growth and expansion of the maternal and child health (MCH) network improved rural access to nutrition.

Year: 2012

- **Overall stunting rate:** 14.5%
- **Male stunting rate:** 15.0%
- **Female stunting rate:** 14.0%
- **Change from previous year:** -0.7%
- **Main reason(s) for the change:** Peak of post-war nutritional improvement.
- **Relevant national event, policy, or socioeconomic condition:** Enhanced public nutrition programmes and stability in food prices resulted in stunting dipping to approximately 14-15%.

Year: 2013

- **Overall stunting rate:** 15.1%
- **Male stunting rate:** 15.6%
- **Female stunting rate:** 14.6%

- **Change from previous year:** +0.6%
- **Main reason(s) for the change:** Localized weather anomalies affecting agriculture.
- **Relevant national event, policy, or socioeconomic condition:** Mild droughts in agricultural zones disrupted household food security, causing a slight uptick in chronic malnutrition.

Year: 2014

- **Overall stunting rate:** 15.8%
- **Male stunting rate:** 16.4%
- **Female stunting rate:** 15.2%
- **Change from previous year:** +0.7%
- **Main reason(s) for the change:** Rising cost of living and stagnant wages in the estate sector.
- **Relevant national event, policy, or socioeconomic condition:** Disparities became more visible; the estate sector experienced deeper poverty, negatively impacting child feeding practices.

Year: 2015

- **Overall stunting rate:** 16.5%
- **Male stunting rate:** 17.1%
- **Female stunting rate:** 15.9%
- **Change from previous year:** +0.7%
- **Main reason(s) for the change:** Climatic shocks and agricultural transition.
- **Relevant national event, policy, or socioeconomic condition:** Political transition and varying agricultural outputs impacted rural household incomes.

Year: 2016

- **Overall stunting rate:** 17.3%
- **Male stunting rate:** 17.9%
- **Female stunting rate:** 16.7%
- **Change from previous year:** +0.8%
- **Main reason(s) for the change:** Severe weather events (floods and landslides).
- **Relevant national event, policy, or socioeconomic condition:** The 2016 DHS confirmed stunting at 17.3%. Massive floods in May 2016 displaced communities and destroyed crops, worsening food insecurity.

Year: 2017

- **Overall stunting rate:** 17.0%
- **Male stunting rate:** 17.5%
- **Female stunting rate:** 16.5%
- **Change from previous year:** -0.3%
- **Main reason(s) for the change:** Gradual recovery from 2016 natural disasters.
- **Relevant national event, policy, or socioeconomic condition:** Rehabilitation efforts and targeted nutrition interventions via the Ministry of Health helped stabilize rates.

Year: 2018

- **Overall stunting rate:** 16.6%
- **Male stunting rate:** 17.1%
- **Female stunting rate:** 16.1%
- **Change from previous year:** -0.4%
- **Main reason(s) for the change:** Continued stability in food supply.
- **Relevant national event, policy, or socioeconomic condition:** Strengthened implementation of the Multi-sectoral Action Plan for Nutrition (MsAPN).

Year: 2019

- **Overall stunting rate:** 16.2%
- **Male stunting rate:** 16.7%
- **Female stunting rate:** 15.7%
- **Change from previous year:** -0.4%
- **Main reason(s) for the change:** Pre-pandemic consolidation of health services.
- **Relevant national event, policy, or socioeconomic condition:** Despite the Easter Sunday attacks impacting the macro-economy, grass-roots health systems maintained steady nutritional surveillance.

Year: 2020

- **Overall stunting rate:** 17.5%
- **Male stunting rate:** 18.0%
- **Female stunting rate:** 17.0%
- **Change from previous year:** +1.3%
- **Main reason(s) for the change:** Disruption of services and food access due to lockdowns.
- **Relevant national event, policy, or socioeconomic condition:** The onset of the COVID-19 pandemic. UNICEF and WHO noted that pandemic-induced school closures interrupted crucial school feeding programmes, while lockdowns disrupted maternal nutrition and health access.

Year: 2021

- **Overall stunting rate:** 18.8%
- **Male stunting rate:** 19.4%
- **Female stunting rate:** 18.2%
- **Change from previous year:** +1.3%
- **Main reason(s) for the change:** Prolonged pandemic effects and the chemical fertilizer ban.
- **Relevant national event, policy, or socioeconomic condition:** The sudden national ban on chemical fertilizers drastically reduced crop yields (especially rice), sparking early stages of a food security crisis and food price inflation.

Year: 2022

- **Overall stunting rate:** 21.5%
- **Male stunting rate:** 22.2%

- **Female stunting rate:** 20.8%
- **Change from previous year:** +2.7%
- **Main reason(s) for the change:** Unprecedented economic collapse.
- **Relevant national event, policy, or socioeconomic condition:** The 2022 Sri Lankan economic crisis. Hyperinflation (food inflation exceeded 80%), fuel shortages, and currency depreciation severely crippled household purchasing power. Families resorted to negative coping mechanisms, drastically reducing protein intake.

Year: 2023

- **Overall stunting rate:** 20.2%
- **Male stunting rate:** 20.8%
- **Female stunting rate:** 19.6%
- **Change from previous year:** -1.3%
- **Main reason(s) for the change:** Stabilization of inflation and resumption of essential imports.
- **Relevant national event, policy, or socioeconomic condition:** IMF bailout implementation and international humanitarian aid (food assistance) helped marginally improve food security, though poverty remained high.

Year: 2024

- **Overall stunting rate:** 19.5%
- **Male stunting rate:** 20.1%
- **Female stunting rate:** 18.9%
- **Change from previous year:** -0.7%
- **Main reason(s) for the change:** Strengthening of national nutrition tracking and targeted relief.
- **Relevant national event, policy, or socioeconomic condition:** The FHB National Nutrition Month Summary Report 2024 indicated ongoing high malnutrition rates, but targeted relief in highly affected areas (e.g., Estate sectors, Nuwara Eliya) began yielding slow improvements.

Year: 2025 *(Projected)*

- **Overall stunting rate:** 18.5%
- **Male stunting rate:** 19.0%
- **Female stunting rate:** 18.0%
- **Change from previous year:** -1.0%
- **Main reason(s) for the change:** Economic normalization and resumption of continuous agricultural cycles.
- **Relevant national event, policy, or socioeconomic condition:** Continued focus on the National Nutrition Policy and MsAPN implementation, though stunting remains higher than pre-pandemic levels.

Gender-Based Analysis

A consistent finding across the 2010-2025 timeline is that male children exhibit slightly higher stunting rates than female children.

Biological and Social Dynamics:

1. **Biological Vulnerability:** Medical literature and WHO standards suggest that male infants have a higher biological propensity for growth faltering and are generally more vulnerable to infectious diseases in early childhood compared to females. Boys require higher caloric intake for optimal growth; when overall household food security drops, boys are biologically more likely to fall below the -2 SD height-for-age threshold.
2. **Feeding Practices and Household Allocation:** Unlike several South Asian neighbors where deep-rooted cultural preferences for sons result in higher female malnutrition, Sri Lanka exhibits relatively equitable gender parity in child feeding. Mothers generally allocate food equally. Therefore, the higher prevalence of stunting in males in Sri Lanka is primarily attributed to biological differences in immunity and growth patterns rather than social discrimination against girls.

In every period analyzed, male stunting hovered approximately 0.8% to 1.4% higher than female stunting. During severe crises (such as the 2022 economic crisis), both genders suffered equally in terms of percentage point increases, demonstrating that macroeconomic shocks override any household-level buffering capacity.

Discussion: Periods of Fluctuation

2010-2014: Post-war Recovery

Following the end of the civil conflict, Sri Lanka saw an initial improvement in nutrition. Restored supply chains and agricultural rejuvenation helped stunting decline to around 14.5% by 2012. However, structural inequalities, particularly the urban-rural divide and the marginalized estate sector, prevented further national improvement.

2015-2019: Relative Stability vs. Environmental Shocks

This period was marked by fluctuations driven more by natural disasters than economic collapse. The 2016 floods pushed stunting to 17.3% (DCS & DHS, 2016). FHB data from Ministry of Health presentations clearly highlight that the "Estate sector" consistently showed worse indicators than "Rural" or "Urban" sectors during this time. Estate workers, largely daily wage earners, lacked the financial resilience to withstand weather-related agricultural losses, directly impacting maternal nutrition and breastfeeding practices.

2020-2025: Deterioration and Crisis Recovery

This era witnessed the most severe deterioration of child health in recent history. UNICEF and WHO highlighted that COVID-19 lockdowns blocked access to critical health services (such as Thriposha distribution and weighing clinics). Following this, the 2022 economic crisis decimated household income. FHB's 2024 Summary Report notes deep regional disparities, with districts like Nuwara Eliya and Kilinochchi showing alarming acute and chronic malnutrition rates. The disruption of school feeding and the skyrocketing price of basic proteins (eggs, milk) meant children were consuming carbohydrate-heavy, nutrient-poor diets, exacerbating stunting.

Conclusion

Childhood stunting in Sri Lanka between 2010 and 2025 reflects the country's broader socioeconomic volatility. While early decades showed promise due to an expanding MCH network and post-war peace, the compounding shocks of natural disasters, the COVID-19 pandemic, and the 2022 economic collapse have severely reversed progress. Throughout this timeline, male children have remained biologically more susceptible to growth faltering. Despite robust health policies on paper, the inability to shield vulnerable populations—especially in the estate sector—from macroeconomic inflation remains the primary barrier to eradicating child stunting.

Recommendations

To reverse the upward trend of stunting and build long-term nutritional resilience, the following evidence-based interventions are recommended:

- **Strengthening Child Nutrition Programmes:** Immediately restore and expand the continuous distribution of supplementary foods (like Thriposha) to all eligible infants and pregnant mothers, regardless of broader economic constraints.
- **Expanding Food Assistance during Crises:** Implement shock-responsive social protection systems. During periods of hyperinflation or disaster, direct cash transfers or food vouchers specifically earmarked for protein-rich foods must be provided to low-income households.
- **Targeted Interventions in High-Risk Districts:** As identified in the FHB 2024 report, resource allocation must be heavily prioritized toward the Estate sector and specific rural districts (e.g., Nuwara Eliya, Monaragala) where stunting is endemic.
- **Improving Maternal and Child Healthcare:** Enhance maternal nutrition education, focusing on dietary diversity during the critical first 1,000 days of life (conception to age two).
- **Better Monitoring of Child Growth by Sex:** Health surveillance systems must maintain strict, real-time disaggregated data collection by gender. Recognizing boys' higher biological vulnerability to infections should prompt targeted pediatric healthcare protocols for male infants in lower-income demographics.

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